

# Jihong Min — List of Publications

Presidential Young Professor, Department of Biomedical Engineering

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Google Scholar

34 papers (9 as first/co-first author), 9,897 total citations, h-index 25, updated August 2026.

† indicates equal contributions

## Selected Publications

1. **Min, J.†**, Ahn, H.†, Lukas, H., Ma, X., Bhansali, R., Sunwoo, S.-H., Wang, C., Xu, Y., Yao, R., Kim, G., Li, Z., Hsiai, T. K., Emami, A., Jung, H.-T., & Gao, W. (2025). Continuous biochemical profiling of the gastrointestinal tract using a multiparametric ingestible capsule. *Nature Electronics*, 8, 844-855. **Featured on Journal Cover (62 citations)**
2. **Min, J.†**, Demchyshyn, S.†, Sempionatto, J. R., Song, Y., Hailegnaw, B., Xu, C., Yang, Y., Solomon, S., Putz, C., Lehner, L., Schwarz, J. F., Schwarzing, C., Scharber, M., Shirzaei Sani, E., Kaltenbrunner, M., & Gao, W. (2023). An autonomous wearable biosensor powered by a perovskite solar cell. *Nature Electronics*, 6, 630-641. **Featured on Journal Cover (395 citations)**
3. Wang, M.†, Yang, Y.†, **Min, J.†**, Song, Y., Tu, J., Mukasa, D., Ye, C., Xu, C., Heflin, N., McCune, J. S., Hsiai, T. K., Li, Z., & Gao, W. (2022). A wearable electrochemical biosensor for the monitoring of metabolites and nutrients. *Nature Biomedical Engineering*, 6, 1225-1235. **Featured on Journal Cover (971 citations)**
4. Song, Y.†, **Min, J.†**, Yu, Y., Wang, H., Yang, Y., Zhang, H., & Gao, W. (2020). Wireless battery-free wearable sweat sensor powered by human motion. *Science Advances*, 6(40), eaay9842. (706 citations)
5. **Min, J.†**, Tu, J.†, Xu, C.†, Lukas, H.†, Shin, S., Yang, Y., Solomon, S. A., Mukasa, D., & Gao, W. (2023). Skin-interfaced wearable sweat sensors for precision medicine. *Chemical Reviews*, 123, 5049-5138. **Featured on Journal Cover (739 citations)**
6. Heng, W., Yin, S., **Min, J.**, Wang, C., Han, H., Shirzaei Sani, E., Li, J., Song, Y., Rossiter, H. B., & Gao, W. (2024). A smart mask for exhaled breath condensate harvesting and analysis. *Science*, 385, 954-961. (214 citations)
7. Tu, J., **Min, J.**, Song, Y., Xu, C., Li, J., Moore, J., Hanson, J., Hu, E., Parimon, T., Wang, T.-Y., Davoodi, E., Chou, T.-F., Chen, P., Hsu, J. J., Rossiter, H. B., & Gao, W. (2023). A wireless patch for the monitoring of C-reactive protein in sweat. *Nature Biomedical Engineering*, 7, 1293-1306. (429 citations)
8. **Min, J.**, Song, Y., & Gao, W. (2022). Microcracked conductors for wearable sensors. *Nature Electronics*, 5(11), 717-718. (10 citations)
9. **Min, J.†**, Sempionatto, J. R.†, Teymourian, H.†, Wang, J., & Gao, W. (2021). Wearable electrochemical biosensors in North America. *Biosensors and Bioelectronics*, 172, 112750. (304 citations)
10. **Min, J.**, Yang, Y., Wu, Z., & Gao, W. (2020). Robotics in the gut. *Advanced Therapeutics*, 3(4), 1900125. (110 citations)

## Other Publications

1. Kim, I., Yao, D. R., Yin, S., Tran-Ngoc, P. T., **Min, J.**, Ng, N., Heng, W., Li, J., Sadri, B., Sato, H., Kim, S. O., & Gao, W. (2026). BeetleBot: An integrated bioinspired soft robot for multimodal sensing and adaptive interaction. *Science Advances*, 12(34), eaeg8849. (0 citations)
2. Xu, C., Zheng, H., Heng, W., Liu, R., MarionSims, J., Li, J., Jin, P., Tay, R. Y., **Min, J.**, Wang, G., Yue, Y., & Gao, W. (2026). Human-inspired time-series health evaluation with an adaptive multimodal electronic skin. *Science Advances*, 12(31), eaeg5606. (0 citations)
3. Kim, G., Cho, S., Ahn, H., Wang, C., Han, H., Ma, X., Kim, M.-J., Qu, J., **Min, J.**, & Gao, W. (2026). Strain-isolated microneedles for ambulatory hormonal and metabolic monitoring. *Nature Nanotechnology*. (0 citations)
4. Gao, B., Ge, Z., Sun, X., Lin, B., Gao, Y., **Min, J.**, Zhang, Q., Rogers, J. A., Yu, X., Gao, W., & Lim,

- C. T. (2026). AI-powered closed-loop wearable bioelectronics for personalized and autonomous healthcare. *Nature Sensors*, 1, 667-680. (1 citation)
5. Li, J., Fan, K., Ma, X., Wang, C., **Min, J.**, Chen, Y., Song, Y., Xu, Y., & Gao, W. (2026). Strain-insensitive wet-tissue-adhesive biphasic bioelectronics for physicochemical monitoring and adaptive therapy. *Nature Materials*. (4 citations)
  6. Xu, Y., Ma, X., Fan, K., **Min, J.**, Qu, J., Heng, W., Li, J., Tang, S., Kim, G., Liu, R., Yin, S., Li, W., & Gao, W. (2026). Strain-resilient intrinsically stretchable electrochemical biointerfaces. *Science*, 392(6801), ead1630. (4 citations)
  7. Heng, W., Putz, C., Tang, W., **Min, J.**, Lehner, L., Demchyshyn, S., Kim, G., Wang, C., Hailegnaw, B., Rippatha, E., Scharber, M., Kim, M., Chen, Y., Fan, Z., Kaltenbrunner, M., Gao, W. (2026). A battery-free smart mask for long-term exhaled breath biochemical sensing. *Nature Sensors*, 1, 328-340. (10 citations)
  8. Shin, S., Liu, R., Yang, Y., Lasalde-Ramírez, J. A., Kim, G., Won, C., **Min, J.**, Wang, C., Fan, K., Han, H., Uwakwe, C., Heng, W., Hsiai, T. K., Li, Z., FitzGerald, J. D., & Gao, W. (2025). A bioinspired microfluidic wearable sensor for multiday sweat sampling, transport, and metabolic analysis. *Science Advances*, 11(33), eadw9024. (80 citations)
  9. Wang, C., Fan, K., Shirzaei Sani, E., Lasalde-Ramírez, J. A., Heng, W., **Min, J.**, Solomon, S. A., Wang, M., Li, J., Han, H., Kim, G., Shin, S., Seder, A., Shih, C.-D., Armstrong, D. G., & Gao, W. (2025). A microfluidic wearable device for wound exudate management and continuous in situ analysis of wound biomarkers. *Science Translational Medicine*, 17(795), eadt0882. (101 citations)
  10. Tu, J., Yeom, J., Chaj Ulloa, J., Solomon, S. A., **Min, J.**, Heng, W., Kim, G., Dao, J., Vemu, R., Pang, M., Wang, C., Kim, D.-H., & Gao, W. (2025). Stressomic: A wearable microfluidic biosensor for dynamic profiling of multiple stress hormones in sweat. *Science Advances*, 11(32), eadx6491. (77 citations)
  11. Wang, M., Ye, C., Yang, Y., Mukasa, D., Wang, C., Xu, C., **Min, J.**, Solomon, S. A., Tu, J., Shen, G., Tang, S., McCune, J. S., Hsiai, T. K., Li, Z., & Gao, W. (2024). Printable molecule-selective core-shell nanoparticles for wearable and implantable sensing. *Nature Materials*, 24, 589-598. (118 citations)
  12. Xu, C., Song, Y., Sempionatto, J. R., Solomon, S. A., Yu, Y., Nyein, H. Y. Y., Tay, R. Y., Li, J., Heng, W., **Min, J.**, Lao, A., Hsiai, T. K., Sumner, J. A., & Gao, W. (2024). A physicochemical-sensing electronic skin for stress response monitoring. *Nature Electronics*, 7, 168-179. (422 citations)
  13. Ye, C., Wang, M., **Min, J.**, Tay, R. Y., Lukas, H., Sempionatto, J. R., Li, J., Xu, C., & Gao, W. (2024). A wearable aptamer nanobiosensor for non-invasive female hormone monitoring. *Nature Nanotechnology*, 19, 330-337. (480 citations)
  14. Song, Y., Tay, R. Y., Li, J., Xu, C., **Min, J.**, Shirzaei Sani, E., Kim, G., Heng, W., Kim, I., & Gao, W. (2023). 3D-printed epifluidic electronic skin for machine learning-powered multimodal health surveillance. *Science Advances*, 9(37), eadi6492. (258 citations)
  15. **Min, J.**, Gao, W. (2023). Battery-Free Wearable Electrochemical Sweat Sensors. *IEEE IFETC*. (2 citations)
  16. Choi, Y., Ho, D. H., Kim, S., Choi, Y. J., Roe, D. G., Kwak, I. C., **Min, J.**, Han, H., Gao, W., & Cho, J. H. (2023). Physically defined long-term and short-term synapses for the development of reconfigurable analog-type operators capable of performing health care tasks. *Science Advances*, 9(27), eadg5946. (61 citations)
  17. Shirzaei Sani, E., Xu, C., Wang, C., Song, Y., **Min, J.**, Tu, J., Solomon, S. A., Li, J., Banks, J. L., & Armstrong, D. G. (2023). A stretchable wireless wearable bioelectronic system for multiplexed monitoring and combination treatment of infected chronic wounds. *Science Advances*, 9(12), eadf7388. (437 citations)
  18. Mukasa, D., Wang, M., **Min, J.**, Yang, Y., Solomon, S. A., Han, H., Ye, C., & Gao, W. (2023). A Computationally Assisted Approach for Designing Wearable Biosensors toward Non-invasive Personalized Molecular Analysis. *Advanced Materials*, 35(35), 2212161. (95 citations)
  19. Yu, Y., Li, J., Solomon, S. A., **Min, J.**, Tu, J., Guo, W., Xu, C., Song, Y., & Gao, W. (2022). All-printed soft human-machine interface for robotic physicochemical sensing. *Science Robotics*, 7(67),

eabn0495. **Featured on Journal Cover** (392 citations)

20. Torrente-Rodríguez, R. M., Lukas, H., Tu, J., **Min, J.**, Yang, Y., Xu, C., Rossiter, H. B., & Gao, W. (2020). SARS-CoV-2 RapidPlex: a graphene-based multiplexed telemedicine platform for rapid and low-cost COVID-19 diagnosis and monitoring. *Matter*, 3(6), 1981-1998. **Featured on Journal Cover** (528 citations)
21. Torrente-Rodríguez, R. M., Tu, J., Yang, Y., **Min, J.**, Wang, M., Song, Y., Yu, Y., Xu, C., Ye, C., & IsHak, W. W. (2020). Investigation of cortisol dynamics in human sweat using a graphene-based wireless mHealth system. *Matter*, 2(4), 921-937. (524 citations)
22. Yang, Y., Song, Y., Bo, X., **Min, J.**, Pak, O. S., Zhu, L., Wang, M., Tu, J., Kogan, A., & Zhang, H. (2020). A laser-engraved wearable sensor for sensitive detection of uric acid and tyrosine in sweat. *Nature Biotechnology*, 38(2), 217-224. (1363 citations)
23. Yu, Y., Nassar, J., Xu, C., **Min, J.**, Yang, Y., Dai, A., Doshi, R., Huang, A., Song, Y., & Gehlhar, R. (2020). Biofuel-powered soft electronic skin with multiplexed and wireless sensing for human-machine interfaces. *Science Robotics*, 5(41), eaaz7946. (724 citations)
24. Song, Y., **Min, J.**, & Gao, W. (2019). Wearable and implantable electronics: moving toward precision therapy. *ACS Nano*, 13(11), 12280-12286. (261 citations)